

**Supplementary materials for the paper “Nonparametric
Eigenvalue-Regularized Precision or Covariance Matrix
Estimator”- Proof of Theorem 1, Theorem 3, Lemma 1, Theorem
5 and Theorem 6.**

We state an important Lemma first, which is in fact Lemma 2.7 of Bai and Silverstein (1998). Then, we present two more lemmas before presenting the proof of Theorem 1.

Lemma S.1 *Let $\mathbf{y} = (y_1, \dots, y_p)^\top$ be a complex random vector with independent and identically distributed (i.i.d.) entries satisfying $E(y_1) = 0$, $E|y_1|^2 = 1$. Let $\mathbf{A}$ be a given $p \times p$ matrix with possibly complex entries. Then for any $q \geq 2$,*

$$E|\mathbf{y}^\top \mathbf{A} \mathbf{y} - \text{tr}(\mathbf{A})|^q \leq K_q \left(E^{q/2} |y_1|^{4\text{tr} q/2} (\mathbf{A} \mathbf{A}^*) + E|y_1|^{2q} \text{tr}(\mathbf{A} \mathbf{A}^*)^{q/2} \right),$$

where K_q is a constant depending only on q , and $\mathbf{A}^* = \bar{\mathbf{A}}^\top$ is the Hermitian of $\mathbf{A}$.

Lemma S.2 *For $z \in \mathbb{C}^+$, we have $\|R_1(z)\|^2 \leq \frac{1}{\text{Im}^2(z)}$.*

Proof of Lemma S.2. Denote $\bar{z}$ the conjugate of z , and let $\tilde{\Sigma}_1 = \mathbf{P}_1 \mathbf{D}_1 \mathbf{P}_1^\top$. Then

$$\begin{aligned} \|R_1(z)\|^2 &= \lambda_{\max}(R_1(z) \overline{R_1(z)}) \\ &= \lambda_{\max}(\mathbf{P}_1 (\mathbf{D}_1 - z \mathbf{I}_p)^{-1} \mathbf{P}_1^\top \mathbf{P}_1 (\mathbf{D}_1 - \bar{z} \mathbf{I}_p)^{-1} \mathbf{P}_1^\top) \\ &= \lambda_{\max}(\{[\mathbf{D}_1 - \text{Re}(z)]^2 + \text{Im}^2(z)\}^{-1}) \\ &\leq \frac{1}{\text{Im}^2(z)}. \quad \square \end{aligned}$$

Lemma S.3 *Let $\mathbf{R}$ be Hermitian, that is, it is symmetric with complex entries. Then for a real square matrix $\mathbf{B}$, we have $|\text{tr}(\mathbf{B} \mathbf{R})| \leq |\lambda_{\max}(\mathbf{B})| |\text{tr}(\mathbf{R})|$. Also, $|\text{tr}(\mathbf{A}^\top \mathbf{R} \mathbf{A})| \leq 2^{1/2} \|\mathbf{R}\| \text{tr}(\mathbf{A}^\top \mathbf{A})$ for a real matrix $\mathbf{A}$.*

Proof of Lemma S.3. Since $\mathbf{R}$ is Hermitian, it can be decomposed as $\mathbf{R} = \mathbf{U} \mathbf{D} \mathbf{U}^\top$, where $\mathbf{U}$ is orthogonal and $\mathbf{D}$ is diagonal with complex entries. Writing $\mathbf{U} = (\mathbf{u}_1, \dots, \mathbf{u}_p)$ and $\mathbf{D} = \text{diag}(d_1, \dots, d_p)$, then

$$\begin{aligned} |\text{tr}(\mathbf{B} \mathbf{R})| &= |\text{tr}(\mathbf{U}^\top \mathbf{B} \mathbf{U} \mathbf{D})| = \left| \sum_{i=1}^p d_i \mathbf{u}_i^\top \mathbf{B} \mathbf{u}_i \right| \\ &\leq |\lambda_{\max}(\mathbf{B})| \cdot \left| \sum_{i=1}^p d_i \right| = |\lambda_{\max}(\mathbf{B})| |\text{tr}(\mathbf{R})|. \end{aligned}$$

$$\begin{aligned} E(|I_1|^6 | \mathbf{Y}_1) &= E \left(\left| \frac{1}{pn_2} \text{tr} \left(R_1(z) \sum_{k=1}^{n_2} \mathbf{y}_{2k} \mathbf{y}_{2k}^T \right) - \frac{1}{p} \text{tr}(R_1(z) \Sigma_p) \right|^6 \middle| \mathbf{Y}_1 \right) \\ &= E \left(\left| \frac{1}{pn_2} \sum_{k=1}^{n_2} (\mathbf{z}_{2k}^T \Sigma_p^{1/2} R_1(z) \Sigma_p^{1/2} \mathbf{z}_{2k} - \text{tr}(R_1(z) \Sigma_p)) \right|^6 \middle| \mathbf{Y}_1 \right). \end{aligned}$$

Define $g_k = \mathbf{z}_{2k}^\top \Sigma_p^{1/2} R_1(z) \Sigma_p^{1/2} \mathbf{z}_{2k} - \text{tr}(R_1(z) \Sigma_p)$. Then $E(g_i | \mathbf{Y}_1) = 0$, and $g_i | \mathbf{Y}_1$ is independent of $g_j | \mathbf{Y}_1$ for $i \neq j$. We then have the expansion

$$\begin{aligned}
E(|I_1|^6) &= \frac{1}{p^6 n_2^6} \sum_{i_1, \dots, i_6=1}^{n_2} E(E(g_{i_1} \bar{g}_{i_2} g_{i_3} \bar{g}_{i_4} g_{i_5} \bar{g}_{i_6} | \mathbf{Y}_1)) \\
&\leq \frac{1}{p^6 n_2^6} E \left(\sum_{i=1}^{n_2} E(|g_i|^6 | \mathbf{Y}_1) + \sum_{i \neq j} [E(|g_i|^2 | \mathbf{Y}_1) E(|g_j|^4 | \mathbf{Y}_1) \right. \\
&\quad + E(|g_i|^3 | \mathbf{Y}_1) E(|g_j|^3 | \mathbf{Y}_1)] \\
&\quad \left. + \sum_{i \neq j \neq k} E(|g_i|^2 | \mathbf{Y}_1) E(|g_j|^2 | \mathbf{Y}_1) E(|g_k|^2 | \mathbf{Y}_1) \right) \\
(S.1) \quad &= O(p^{-6} n_2^{-6} \cdot [n_2 + 2n_2(n_2 - 1) + n_2(n_2 - 1)(n_2 - 2)] p^6) = O(n_2^{-3}),
\end{aligned}$$

where the last line used $E(|g_i|^q | \mathbf{Y}_1) = O(p^q)$ proved before. Since $\sum_{n \geq 1} n_2^{-3} < \infty$ by assumption, we can apply the Borel-Cantelli lemma to conclude that $I_1 \xrightarrow{a.s.} 0$. This completes the proof of part (i).

Note also that since $I_1 \xrightarrow{a.s.} 0$, we have $\Psi_m^{(1)}(z) - p^{-1}\text{tr}(R_1(z)\Sigma_p) \xrightarrow{a.s.} 0$. Using the inversion formula (2.3), it is easy to show that the inverse Stieltjes transform for $\Psi_m^{(1)}(z)$ is $\Phi_m^{(1)}(x)$ in (2.13) on all points of continuity of $\Phi_m^{(1)}(x)$, and that for $p^{-1}\text{tr}(R_1(z)\Sigma_p)$ is $p^{-1} \sum_{i=1}^p \mathbf{p}_{1i}^\top \Sigma_p \mathbf{p}_{1i} \mathbf{1}_{\{\lambda_{1i} \leq x\}}$ on all points of continuity of the function. Hence, using equation (2.5) of Silverstein and Bai (1995), we conclude that

$$\Phi_m^{(1)}(x) - \frac{1}{p} \sum_{i=1}^p \mathbf{p}_{1i}^T \Sigma_p \mathbf{p}_{1i} \mathbf{1}_{\{\lambda_{1i} \leq x\}} \xrightarrow{a.s.} 0$$

as $n_1, p \rightarrow \infty$ with $p/n_1 \rightarrow c_1 > 0$, which certainly include the case $c_1 = 1$. This proves the last part of the theorem.

The proof for part (ii) of the theorem can be found in the proof of Theorem 4 of Ledoit and P  ch   (2011), from equations (45) to (50) by replacing γ there with $1/c_1$ and F with F_1 . Part (iii) is just an application of Theorem 4 of Ledoit and P  ch   (2011). This completes the proof of the theorem. $\square$

Proof of Theorem 3. Let $\mathbf{Y}_2 = (\mathbf{y}_{21}, \dots, \mathbf{y}_{2n_2})$ with $\mathbf{y}_{2k} = \mathbf{A}\mathbf{x}_{2k} + \boldsymbol{\epsilon}_{2k}$, $k = 1, \dots, n_2$. Let $R_1(z) = (\tilde{\boldsymbol{\Sigma}}_1 - z\mathbf{I}_p)^{-1}$. Then with $\boldsymbol{\Sigma}_p = \mathbf{A}\boldsymbol{\Sigma}_x\mathbf{A} + \boldsymbol{\Sigma}_\epsilon$ as

in (3.2), we have

$$\begin{aligned} \frac{1}{p} \text{tr}(R_1(z) \tilde{\Sigma}_2) - \frac{1}{p} \text{tr}(R_1(z) \Sigma_p) &= D_1 + D_2 + D_3, \text{ where} \\ D_1 &= \frac{1}{pn_2} \sum_{k=1}^{n_2} \{ \mathbf{x}_{2k}^T \mathbf{A}^T R_1(z) \mathbf{A} \mathbf{x}_{2k} - \text{tr}(R_1(z) \mathbf{A} \Sigma_x \mathbf{A}^T) \}, \\ D_2 &= \frac{2}{pn_2} \sum_{k=1}^{n_2} \mathbf{x}_{2k}^T \mathbf{A}^T R_1(z) \boldsymbol{\epsilon}_{2k}, \\ D_3 &= \frac{1}{pn_2} \sum_{k=1}^{n_2} \{ \boldsymbol{\epsilon}_{2k}^T R_1(z) \boldsymbol{\epsilon}_{2k} - \text{tr}(R_1(z) \Sigma_\epsilon) \}. \end{aligned}$$

Consider D_1 first. Using assumption (F1), and defining

$$g_i = \mathbf{x}_{2k}^{*T} \Sigma_x^{1/2} \mathbf{A}^T R_1(z) \mathbf{A} \Sigma_x^{1/2} \mathbf{x}_{2k}^* - \text{tr}(R_1(z) \mathbf{A} \Sigma_x \mathbf{A}^T),$$

for $2 \leq q \leq 6$ and $z \in \mathbb{C}^+$, we have by Lemma S.1,

$$\begin{aligned} E(|g_k|^q | \mathbf{Y}_1) &\leq K_q (E^{q/2} |x_{2k}^*|^4 \text{tr}^{q/2}(\Sigma_x^{1/2} \mathbf{A}^T R_1(z) \mathbf{A} \Sigma_x \mathbf{A}^T \overline{R_1(z)} \mathbf{A} \Sigma_x^{1/2}) \\ &\quad + E |x_{2k}^*|^{2q} \text{tr}(\Sigma_x^{1/2} \mathbf{A}^T R_1(z) \mathbf{A} \Sigma_x \mathbf{A}^T \overline{R_1(z)} \mathbf{A} \Sigma_x^{1/2})^{q/2}) \\ &\leq K_q \|\Sigma_x^{1/2} \mathbf{A}^T R_1(z) \mathbf{A} \Sigma_x \mathbf{A}^T \overline{R_1(z)} \mathbf{A} \Sigma_x^{1/2}\|^{q/2} \\ &\quad \cdot (E^{q/2} |x_{2k}^*|^4 r^{q/2} + r E |x_{2k}^*|^{2q}) \\ &\leq K_q \|\Sigma_x\|^q \|\mathbf{A}\|^{2q} \text{Im}^{-q}(z) (E^{q/2} |x_{2k}^*|^4 r^{q/2} + r E |x_{2k}^*|^{2q}) = O(p^q), \end{aligned}$$

where we have used $\|\mathbf{A}\| = O(p^{1/2})$ implied in assumption (F2). Using the fact that $E(g_k | \mathbf{Y}_1) = 0$ and $g_i | \mathbf{Y}_1$ is independent of $g_j | \mathbf{Y}_1$ for $i \neq j$ by assumption (F1), similar to the expansion in (S.1), we have

$$E(|D_1|^6) = \frac{1}{p^6 n_2^6} \sum_{i_1 \dots i_6=1}^{n_2} E(E(g_{i_1} \bar{g}_{i_2} g_{i_3} \bar{g}_{i_4} g_{i_5} \bar{g}_{i_6} | \mathbf{Y}_1)) = O(n_2^{-3}),$$

so that $D_1 \xrightarrow{a.s.} 0$ by the Borel-Cantelli lemma, as $\sum_{n \geq 1} n_2^{-3} < \infty$ by assumption.

Now consider D_2 . Define $g_k = \mathbf{x}_{2k}^T \mathbf{A}^T R_1(z) \boldsymbol{\epsilon}_{2k}$. We have for $0 < q \leq 12$ and $z \in \mathbb{C}^+$,

$$E(|g_k|^q | \mathbf{Y}_1) \leq E \|\mathbf{x}_{2k}^*\|^q \|\Sigma_x\|^{q/2} \|\mathbf{A}\|^q E \|\boldsymbol{\xi}_{2k}\|^q \|\Sigma_\epsilon\|^{q/2} \text{Im}^{-q}(z) = O(p^q),$$

by assumptions (F1) and (F2), and lemma S.2. Hence, using the fact that $E(g_k | \mathbf{Y}_1) = 0$ and $g_i | \mathbf{Y}_1$ is independent of $g_j | \mathbf{Y}_1$ for $i \neq j$, an expansion similar to (S.1) leads again to

$$E(|D_2|^6) = O(n_2^{-3}),$$

so that by the Borel-Cantelli lemma, we have $D_2 \xrightarrow{a.s.} 0$.

Finally, if we define $g_k = \boldsymbol{\xi}_{2k}^\top \boldsymbol{\Sigma}_\epsilon^{1/2} R_1(z) \boldsymbol{\Sigma}_\epsilon^{1/2} \boldsymbol{\xi}_{2k} - \text{tr}(R_1(z) \boldsymbol{\Sigma}_\epsilon)$, then following exactly the same arguments as in the proof of Theorem 1, we can conclude

$$E(|D_3|^6) = O(n_2^{-3}),$$

so that through the Borel-Cantelli lemma we have $D_3 \xrightarrow{a.s.} 0$.

Hence, we have proved

$$\frac{1}{p} \text{tr}[R_1(z) \tilde{\boldsymbol{\Sigma}}_2] - \frac{1}{p} \text{tr}[R_1(z) \boldsymbol{\Sigma}_p] \xrightarrow{a.s.} 0.$$

Then, using equation (2.5) of Silverstein and Bai (1995), we conclude the almost sure convergence of the inverse Stieltjes transform on all points of continuity x , that is,

$$\frac{1}{p} \sum_{i=1}^p \mathbf{p}_{1i}^\top \tilde{\boldsymbol{\Sigma}}_2 \mathbf{p}_{1i} \mathbf{1}_{\{\lambda_{1i} \leq x\}} - \frac{1}{p} \sum_{i=1}^p \mathbf{p}_{1i}^\top \boldsymbol{\Sigma}_p \mathbf{p}_{1i} \mathbf{1}_{\{\lambda_{1i} \leq x\}} \xrightarrow{a.s.} 0. \quad \square$$

Proof of Lemma 1. We first assume (A1)' and (A2)'. For $j = 1, \dots, n_2$ and $i = 1, \dots, p$, define

$$g_{ij} = \frac{\mathbf{z}_{2j}^\top \boldsymbol{\Sigma}_p^{1/2} \mathbf{p}_{1i} \mathbf{p}_{1i}^\top \boldsymbol{\Sigma}_p^{1/2} \mathbf{z}_{2j} - \text{tr}(\boldsymbol{\Sigma}_p^{1/2} \mathbf{p}_{1i} \mathbf{p}_{1i}^\top \boldsymbol{\Sigma}_p^{1/2})}{\mathbf{p}_{1i}^\top \boldsymbol{\Sigma}_p \mathbf{p}_{1i}}.$$

Then for $2 \leq q \leq 10$, using Lemma S.1 and assumption (A1)',

$$\begin{aligned} E(|g_{ij}|^q | \mathbf{Y}_1) &\leq (\mathbf{p}_{1i}^\top \boldsymbol{\Sigma}_p \mathbf{p}_{1i})^{-q} K_q \left(E^{q/2} |z_{2j}|^4 \text{tr}^q(\boldsymbol{\Sigma}_p^{1/2} \mathbf{p}_{1i} \mathbf{p}_{1i}^\top \boldsymbol{\Sigma}_p^{1/2}) \right. \\ &\quad \left. + E |z_{2j}|^{2q} \text{tr}(\boldsymbol{\Sigma}_p^{1/2} \mathbf{p}_{1i} \mathbf{p}_{1i}^\top \boldsymbol{\Sigma}_p^{1/2})^q \right) \\ &= K_q (E^{q/2} |z_{2j}|^4 + E |z_{2j}|^{2q}) = O(1). \end{aligned}$$

At the same time, $E(g_{ij} | \mathbf{Y}_1) = 0$ and $g_{i_1 j_1} | \mathbf{Y}_1$ is independent of $g_{i_2 j_2} | \mathbf{Y}_1$

whenever $j_1 \neq j_2$. Hence,

$$\begin{aligned}
& E \left\{ \max_{1 \leq i \leq p} \left| \frac{\mathbf{p}_{1i}^T \tilde{\Sigma}_2 \mathbf{p}_{1i} - \mathbf{p}_{1i}^T \Sigma_p \mathbf{p}_{1i}}{\mathbf{p}_{1i}^T \Sigma_p \mathbf{p}_{1i}} \right| \right\}^{10} = E \left\{ \max_{1 \leq i \leq p} \left| \frac{1}{n_2} \sum_{j=1}^{n_2} g_{ij} \right| \right\}^{10} \\
& \leq \sum_{i=1}^p \frac{1}{n_2^{10}} \sum_{j_1 \cdots j_{10}=1}^{n_2} E(E \{g_{ij_1} \cdots g_{ij_{10}} | \mathbf{Y}_1\}) \\
& \leq E \left\{ \sum_{i=1}^p \frac{1}{n_2^{10}} \left(\sum_{j=1}^{n_2} E(g_{ij}^{10} | \mathbf{Y}_1) + \sum_{\substack{q_1, q_2 \neq 1 \\ q_1 + q_2 = 10}} \sum_{j_1 \neq j_2} E(g_{ij_1}^{q_1} | \mathbf{Y}_1) E(g_{ij_2}^{q_2} | \mathbf{Y}_1) \right. \right. \\
& \quad + \sum_{\substack{q_1, q_2, q_3 \neq 1 \\ q_1 + q_2 + q_3 = 10}} \sum_{j_1 \neq j_2 \neq j_3} \prod_{k=1}^3 E(g_{ij_k}^{q_k} | \mathbf{Y}_1) + \sum_{\substack{q_1, \dots, q_4 \neq 1 \\ q_1 + \dots + q_4 = 10}} \sum_{j_1 \neq j_2 \neq j_3 \neq j_4} \prod_{k=1}^4 E(g_{ij_k}^{q_k} | \mathbf{Y}_1) \\
& \quad \left. \left. + \sum_{j_1 \neq \dots \neq j_5} \prod_{k=1}^5 E(g_{ij_k}^2 | \mathbf{Y}_1) \right) \right\} \\
& = O(pn_2^{-10} [n_2 + 4P_2^{n_2} + 4P_3^{n_2} + 2P_4^{n_2} + P_5^{n_2}]) \\
& \quad (S.2) \\
& = O(pn_2^{-5}),
\end{aligned}$$

where $P_r^{n_2} = n_2(n_2-1) \cdots (n_2-r+1)$. Since $\sum_{n \geq 1} pn_2^{-5} < \infty$ by assumption, the Borel-Cantelli lemma completes the proof of this part.

Now consider data from a factor model, with assumption (F1)' satisfied. We can decompose

$$\begin{aligned}
& \frac{\mathbf{p}_{1i}^T \tilde{\Sigma}_2 \mathbf{p}_{1i} - \mathbf{p}_{1i}^T \Sigma_p \mathbf{p}_{1i}}{\mathbf{p}_{1i}^T \Sigma_p \mathbf{p}_{1i}} = D_1 + D_2 + D_3, \quad \text{where} \\
& D_1 = \frac{1}{n_2} \sum_{j=1}^{n_2} g_{ij}, \quad g_{ij} = \frac{\mathbf{x}_{2j}^{*T} \Sigma_x^{1/2} \mathbf{A}^T \mathbf{p}_{1i} \mathbf{p}_{1i}^T \mathbf{A} \Sigma_x^{1/2} \mathbf{x}_{2j}^* - \mathbf{p}_{1i}^T \mathbf{A} \Sigma_x \mathbf{A}^T \mathbf{p}_{1i}}{\mathbf{p}_{1i}^T \Sigma_p \mathbf{p}_{1i}}, \\
& D_2 = \frac{1}{n_2} \sum_{j=1}^{n_2} d_{ij}, \quad d_{ij} = \frac{\boldsymbol{\xi}_{2j}^T \Sigma_\epsilon^{1/2} \mathbf{p}_{1i} \mathbf{p}_{1i}^T \Sigma_\epsilon^{1/2} \boldsymbol{\xi}_{2j} - \mathbf{p}_{1i}^T \Sigma_\epsilon \mathbf{p}_{1i}}{\mathbf{p}_{1i}^T \Sigma_p \mathbf{p}_{1i}}, \\
& D_3 = \frac{2}{n_2} \sum_{j=1}^{n_2} h_{ij}, \quad h_{ij} = \frac{\mathbf{x}_{2j}^{*T} \Sigma_x^{1/2} \mathbf{A}^T \mathbf{p}_{1i} \mathbf{p}_{1i}^T \Sigma_\epsilon^{1/2} \boldsymbol{\xi}_{2j}}{\mathbf{p}_{1i}^T \Sigma_p \mathbf{p}_{1i}}.
\end{aligned}$$

Using Lemma S.1, for $2 \leq q \leq 10$,

$$\begin{aligned} E(|g_{ij}|^q | \mathbf{Y}_1) &\leq (\mathbf{p}_{1i}^\top \boldsymbol{\Sigma}_p \mathbf{p}_{1i})^{-q} K_q (\mathbf{p}_{1i}^\top \mathbf{A} \boldsymbol{\Sigma}_x \mathbf{A}^\top \mathbf{p}_{1i})^q (E^{q/2} |x_{2j}^*|^4 + E |x_{2j}^*|^{2q}) \\ &\leq K_q (E^{q/2} |x_{2j}^*|^4 + E |x_{2j}^*|^{2q}) = O(1); \\ E(|d_{ij}|^q | \mathbf{Y}_1) &\leq (\mathbf{p}_{1i}^\top \boldsymbol{\Sigma}_p \mathbf{p}_{1i})^{-q} K_q (\mathbf{p}_{1i}^\top \boldsymbol{\Sigma}_\epsilon \mathbf{p}_{1i})^q (E^{q/2} |\xi_{2j}|^4 + E |\xi_{2j}|^{2q}) \\ &\leq K_q (E^{q/2} |\xi_{2j}|^4 + E |\xi_{2j}|^{2q}) = O(1). \end{aligned}$$

For $4 \leq q \leq 10$, using the c_r -inequality,

$$\begin{aligned} E(|h_{ij}|^q | \mathbf{Y}_1) &= E \left(\left(g_{ij} + \frac{\mathbf{p}_{1i}^\top \mathbf{A} \boldsymbol{\Sigma}_x \mathbf{A}^\top \mathbf{p}_{1i}}{\mathbf{p}_{1i}^\top \boldsymbol{\Sigma}_p \mathbf{p}_{1i}} \right)^{q/2} \middle| \mathbf{Y}_1 \right) \\ &\quad \cdot E \left(\left(d_{ij} + \frac{\mathbf{p}_{1i}^\top \boldsymbol{\Sigma}_\epsilon \mathbf{p}_{1i}}{\mathbf{p}_{1i}^\top \boldsymbol{\Sigma}_p \mathbf{p}_{1i}} \right)^{q/2} \middle| \mathbf{Y}_1 \right), \text{ with} \\ E \left(\left(g_{ij} + \frac{\mathbf{p}_{1i}^\top \mathbf{A} \boldsymbol{\Sigma}_x \mathbf{A}^\top \mathbf{p}_{1i}}{\mathbf{p}_{1i}^\top \boldsymbol{\Sigma}_p \mathbf{p}_{1i}} \right)^{q/2} \middle| \mathbf{Y}_1 \right) &\leq C_{q/2} (E(|g_{ij}|^{q/2} | \mathbf{Y}_1) + 1) = O(1), \\ E \left(\left(\frac{\xi_{2j}^\top \boldsymbol{\Sigma}_\epsilon^{1/2} \mathbf{p}_{1i} \mathbf{p}_{1i}^\top \boldsymbol{\Sigma}_\epsilon^{1/2} \xi_{2j}}{\mathbf{p}_{1i}^\top \boldsymbol{\Sigma}_p \mathbf{p}_{1i}} \right)^{q/2} \middle| \mathbf{Y}_1 \right) &\leq C_{q/2} (E(|d_{ij}|^{q/2} | \mathbf{Y}_1) + 1) = O(1), \end{aligned}$$

where $C_{q/2}$ is a constant. Hence, $E(|h_{ij}|^q | \mathbf{Y}_1) = O(1)$ for $q \geq 4$. For $2 \leq q < 4$, it is easy to see that $E(|h_{ij}|^q | \mathbf{Y}_1) \leq E^{1/2}(|h_{ij}|^{2q} | \mathbf{Y}_1) = O(1)$, and hence $E(|h_{ij}|^q | \mathbf{Y}_1) = O(1)$ for $2 \leq q \leq 10$. Then since $E(g_{ij} | \mathbf{Y}_1) = 0$ and $g_{i_1 j_1} | \mathbf{Y}_1$ is independent of $g_{i_2 j_2} | \mathbf{Y}_1$ whenever $j_1 \neq j_2$, an expansion exactly like (S.2) will show that

$$E \left(\left| \max_{1 \leq i \leq p} |D_1|^{10} \right| \right) = O(p n_2^{-5}),$$

showing that $\max_{1 \leq i \leq p} |D_1| \xrightarrow{a.s.} 0$, since $\sum_{n \geq 1} p n_2^{-5} < \infty$ by assumption. Using the same expansion (S.2) and the arguments above, we can also show that $\max_{1 \leq i \leq p} |D_2|, \max_{1 \leq i \leq p} |D_3| \xrightarrow{a.s.} 0$. This completes the proof of the lemma. $\square$

Before proving Theorem 5, we state and prove a lemma first.

Lemma S.4 *Let assumptions (A1) to (A4) be satisfied. Then assuming $\boldsymbol{\Sigma}_p \neq \sigma^2 \mathbf{I}$,*

$$\frac{\|\widehat{\boldsymbol{\Sigma}}_{\text{Ideal},m} - \boldsymbol{\Sigma}_p\|_F^2}{\|\widehat{\boldsymbol{\Sigma}}_{\text{Ideal}} - \boldsymbol{\Sigma}_p\|_F^2} \xrightarrow{a.s.} 1, \quad \frac{\log \det(\widehat{\boldsymbol{\Sigma}}_{\text{Ideal},m} \boldsymbol{\Sigma}_p^{-1})}{\log \det(\widehat{\boldsymbol{\Sigma}}_{\text{Ideal}} \boldsymbol{\Sigma}_p^{-1})} \xrightarrow{a.s.} 1.$$

Proof of Lemma S.4. Consider the Frobenius loss first. For λ_i the i th largest eigenvalue of $\mathbf{S}_n$ with corresponding eigenvector $\mathbf{p}_i$,

$$\begin{aligned}
p^{-1} \|\widehat{\Sigma}_{\text{Ideal}} - \Sigma_p\|_F^2 &= p^{-1} \text{tr}(\text{diag}(\mathbf{P}^T \Sigma_p \mathbf{P}) - \mathbf{P}^T \Sigma_p \mathbf{P})^2 \\
&= p^{-1} \sum_{i=1}^n (\mathbf{p}_i^T \Sigma_p \mathbf{p}_i)^2 - 2p^{-1} \text{tr}(\mathbf{P}^T \Sigma_p \mathbf{P} \text{diag}(\mathbf{P}^T \Sigma_p \mathbf{P})) \\
&\quad + p^{-1} \text{tr}(\Sigma_p^2) \\
&= p^{-1} \text{tr}(\Sigma_p^2) - p^{-1} \sum_{i=1}^p (\mathbf{p}_i^T \Sigma_p \mathbf{p}_i)^2,
\end{aligned}$$

so that similarly, $p^{-1} \|\hat{\Sigma}_{\text{Ideal},m} - \Sigma_p\|_F^2 = p^{-1} \text{tr}(\Sigma_p^2) - p^{-1} \sum_{i=1}^p (\mathbf{p}_{1i}^T \Sigma_p \mathbf{p}_{1i})^2$. But with assumption (A3), we have

$$p^{-1}\text{tr}(\Sigma_p^2) = p^{-1} \sum_{i=1}^p \tau_{n,i}^2 \xrightarrow{a.s.} \int \tau^2 dH(\tau).$$

Moreover, with the convergence result in Theorem 4 of Ledoit and P  ch   (2011) indicating that (see equation (2.7) as well) for all $x \in \mathbb{R}$, we have

$$\Delta_p(x) = p^{-1} \sum_{i=1}^p \mathbf{p}_i^T \Sigma_p \mathbf{p}_i \mathbf{1}_{\{\lambda_i \leq x\}} \xrightarrow{a.s.} \Delta(x) = \int_{-\infty}^x \delta(\lambda) dF(\lambda),$$

where $F(\cdot)$ is such that

$$F_p(\lambda) = p^{-1} \sum_{i=1}^p \mathbf{1}_{\{\lambda_i \leq \lambda\}} \xrightarrow{a.s.} F(\lambda),$$

we have for any continuous function $g(\cdot)$ over the positive real line,

$$(S.3) \quad p^{-1} \sum_{i=1}^p g(\mathbf{p}_i^T \Sigma_p \mathbf{p}_i) \mathbf{1}_{\{\lambda_i \leq x\}} \xrightarrow{a.s.} \int_{-\infty}^x g(\delta(\lambda)) dF(\lambda).$$

With this, it is easy to see that then

$$p^{-1} \|\widehat{\Sigma}_{\text{Ideal}} - \Sigma_p\|_F^2 \xrightarrow{a.s.} \int \tau^2 dH(\tau) - \int \delta^2(\lambda) dF(\lambda),$$

which is nonzero in general, except when $\Sigma_p = \sigma^2 \mathbf{I}$. At the same time, we have (see equation (2.9) and the descriptions therein as well)

$$p^{-1} \sum_{i=1}^p \mathbf{p}_{1i}^T \Sigma_p \mathbf{p}_{1i} \mathbf{1}_{\{\lambda_{1i} \leq x\}} \xrightarrow{a.s.} \int_{-\infty}^x \delta_1(\lambda) dF_1(\lambda),$$

where $F_1(\cdot)$ is such that

$$F_{1p}(\lambda) = p^{-1} \sum_{i=1}^p \mathbf{1}_{\{\lambda_{1i} \leq \lambda\}} \xrightarrow{a.s.} F_1(\lambda).$$

But since $m/n = n_1/n \rightarrow 1$, we have $p/n_1, p/n$ both going to the same limit $c_1 = c > 0$. Theorem 4.1 of Bai and Silverstein (2010) tells us then both F_p and F_{1p} converges to the same limit almost surely under assumptions (A1) to (A4). Hence $F = F_1$ almost surely, implying $\delta_1(\cdot) = \delta(\cdot)$ almost surely (See Remark 1 in section 2.2 as well). This immediately implies that, similar to (S.3),

$$p^{-1} \sum_{i=1}^p (\mathbf{p}_{1i}^T \Sigma_p \mathbf{p}_{1i})^2 \mathbf{1}_{\{\lambda_{1i} \leq x\}} \xrightarrow{a.s.} \int_{-\infty}^x \delta_1^2(\lambda) dF_1(\lambda) = \int_{-\infty}^x \delta^2(\lambda) dF(\lambda),$$

and hence

$$\begin{aligned} p^{-1} \|\widehat{\Sigma}_{\text{Ideal},m} - \Sigma_p\|_F^2 &\xrightarrow{a.s.} \int \tau^2 dH(\tau) - \int \delta_1^2(\lambda) dF_1(\lambda) \\ &= \int \tau^2 dH(\tau) - \int \delta^2(\lambda) dF(\lambda). \end{aligned}$$

We immediately have

$$\frac{\|\widehat{\Sigma}_{\text{Ideal},m} - \Sigma_p\|_F^2}{\|\widehat{\Sigma}_{\text{Ideal}} - \Sigma_p\|_F^2} \xrightarrow{a.s.} 1,$$

which completes the proof for the Frobenius loss.

For the remaining part, note that it is not difficult to show the inverse Stein's loss

$$\begin{aligned} p^{-1} SL(\Sigma_p, \widehat{\Sigma}_{\text{Ideal}}) &= p^{-1} \log \det(\widehat{\Sigma}_{\text{Ideal}} \Sigma_p^{-1}) \\ &= p^{-1} \sum_{i=1}^p \log(\mathbf{p}_i^T \Sigma_p \mathbf{p}_i) - p^{-1} \sum_{i=1}^p \log(\tau_{n,i}). \end{aligned}$$

Similar arguments as before show that

$$\begin{aligned} p^{-1} \sum_{i=1}^p \log(\mathbf{p}_i^T \Sigma_p \mathbf{p}_i) &\xrightarrow{a.s.} \int \log(\delta(\lambda)) dF(\lambda), \\ p^{-1} \sum_{i=1}^p \log(\tau_{n,i}) &\xrightarrow{a.s.} \int \log(\tau) dH(\tau). \end{aligned}$$

Hence

$$\begin{aligned} p^{-1}SL(\Sigma_p, \widehat{\Sigma}_{\text{Ideal}}) &= p^{-1} \log \det(\widehat{\Sigma}_{\text{Ideal}} \Sigma_p^{-1}) \\ &\xrightarrow{a.s.} \int \log(\delta(\lambda)) dF(\lambda) - \int \log(\tau) dH(\tau), \end{aligned}$$

which is nonzero in general except when $\Sigma_p = \sigma^2 \mathbf{I}$. Moreover, similar arguments as in the case for the Frobenius loss show that

$$\begin{aligned} p^{-1} \sum_{i=1}^p \log(\mathbf{p}_{1i}^T \Sigma_p \mathbf{p}_{1i}) \mathbf{1}_{\{\lambda_{1i} \leq x\}} &\xrightarrow{a.s.} \int_{-\infty}^x \log(\delta_1(\lambda)) dF_1(\lambda) \\ &= \int_{-\infty}^x \log(\delta(\lambda)) dF(\lambda). \end{aligned}$$

Hence,

$$\begin{aligned} p^{-1}SL(\Sigma_p, \widehat{\Sigma}_{\text{Ideal},m}) &= p^{-1} \log \det(\widehat{\Sigma}_{\text{Ideal},m} \Sigma_p^{-1}) \\ &\xrightarrow{a.s.} \int \log(\delta_1(\lambda)) dF_1(\lambda) - \int \log(\tau) dH(\tau) \\ &= \int \log(\delta(\lambda)) dF(\lambda) - \int \log(\tau) dH(\tau). \end{aligned}$$

This completes the proof of the lemma. $\square$

Proof of Theorem 5. We first assume (A1)', (A2)', (A3) and (A4). Since we have

$$\begin{aligned} p^{-1} \|\widehat{\Sigma}_m - \Sigma_p\|_F^2 &= p^{-1} \|\text{diag}(\mathbf{P}_1^T \widetilde{\Sigma}_2 \mathbf{P}_1) - \mathbf{P}_1^T \Sigma_p \mathbf{P}_1\|_F^2 \\ &= p^{-1} \sum_{i=1}^p (\mathbf{p}_{1i}^T \widetilde{\Sigma}_2 \mathbf{p}_{1i} - \mathbf{p}_{1i}^T \Sigma_p \mathbf{p}_{1i})^2 + p^{-1} \|\widehat{\Sigma}_{\text{Ideal},m} - \Sigma_p\|_F^2, \end{aligned}$$

we can express the efficiency loss in (4.2) with respect to the Frobenius loss as

$$\begin{aligned} EL(\Sigma_p, \widehat{\Sigma}_m) &= 1 - \left(\frac{p^{-1} \sum_{i=1}^p (\mathbf{p}_{1i}^T \widetilde{\Sigma}_2 \mathbf{p}_{1i} - \mathbf{p}_{1i}^T \Sigma_p \mathbf{p}_{1i})^2}{p^{-1} \|\widehat{\Sigma}_{\text{Ideal}} - \Sigma_p\|_F^2} \right. \\ &\quad \left. + \frac{p^{-1} \|\widehat{\Sigma}_{\text{Ideal},m} - \Sigma_p\|_F^2}{p^{-1} \|\widehat{\Sigma}_{\text{Ideal}} - \Sigma_p\|_F^2} \right)^{-1}. \end{aligned}$$

With $m/n \rightarrow 1$, we have $p/m \rightarrow c > 0$, same as p/n does. The conclusion from lemma S.4 that the ratio of loss for the ideal estimators being

almost surely 1 then implies that $EL(\hat{\Sigma}_m - \Sigma_p) \xrightarrow{a.s.} 0$, if we can show that $p^{-1} \sum_{i=1}^p (\mathbf{p}_{1i}^T \tilde{\Sigma}_2 \mathbf{p}_{1i} - \mathbf{p}_{1i}^T \Sigma_p \mathbf{p}_{1i})^2 \xrightarrow{a.s.} 0$. But together with assumption $\sum_{n \geq 1} p n_2^{-5} < \infty$, we can apply Lemma 1, so that

$$p^{-1} \sum_{i=1}^p (\mathbf{p}_{1i}^T \tilde{\Sigma}_2 \mathbf{p}_{1i} - \mathbf{p}_{1i}^T \Sigma_p \mathbf{p}_{1i})^2 \leq \left(\max_{1 \leq i \leq p} \left| \frac{\mathbf{p}_{1i}^T \tilde{\Sigma}_2 \mathbf{p}_{1i} - \mathbf{p}_{1i}^T \Sigma_p \mathbf{p}_{1i}}{\mathbf{p}_{1i}^T \Sigma_p \mathbf{p}_{1i}} \right| \right)^2 \cdot \max_{1 \leq i \leq p} (\mathbf{p}_{1i}^T \Sigma_p \mathbf{p}_{1i})^2 \xrightarrow{a.s.} 0,$$

since $\max_{1 \leq i \leq p} (\mathbf{p}_{1i}^T \Sigma_p \mathbf{p}_{1i})^2 \leq \|\Sigma_p\|^2 = O(1)$ by assumption (A2)'. Hence, $EL(\Sigma_p, \hat{\Sigma}_m) \xrightarrow{a.s.} 0$ in this case.

Now consider the inverse Stein's loss. We have

$$\begin{aligned} SL(\Sigma_p, \hat{\Sigma}_m) &= \text{tr}(\mathbf{P}_1^T \Sigma_p \mathbf{P}_1 \text{diag}^{-1}(\mathbf{P}_1^T \tilde{\Sigma}_2 \mathbf{P}_1)) \\ &\quad - \log \det(\mathbf{P}_1^T \Sigma_p \mathbf{P}_1 \text{diag}^{-1}(\mathbf{P}_1^T \tilde{\Sigma}_2 \mathbf{P}_1)) - p \\ &= \sum_{i=1}^p \frac{\mathbf{p}_{1i}^T \Sigma_p \mathbf{p}_{1i}}{\mathbf{p}_{1i}^T \tilde{\Sigma}_2 \mathbf{p}_{1i}} - p + \sum_{i=1}^p \log \left(\frac{\mathbf{p}_{1i}^T \tilde{\Sigma}_2 \mathbf{p}_{1i}}{\mathbf{p}_{1i}^T \Sigma_p \mathbf{p}_{1i}} \right) + SL(\Sigma_p, \hat{\Sigma}_{\text{Ideal}, m}) \\ &= \sum_{i=1}^p h_1(g_i) + \sum_{i=1}^p h_2(g_i) + SL(\Sigma_p, \hat{\Sigma}_{\text{Ideal}, m}), \end{aligned}$$

where we define g_i for $i = 1, \dots, p$, and the functions h_1 and h_2 , which are differentiable on $(-1, \infty)$, to be

$$g_i = \frac{\mathbf{p}_{1i}^T \tilde{\Sigma}_2 \mathbf{p}_{1i} - \mathbf{p}_{1i}^T \Sigma_p \mathbf{p}_{1i}}{\mathbf{p}_{1i}^T \Sigma_p \mathbf{p}_{1i}}, \quad h_1(x) = \frac{1}{1+x} - 1, \quad h_2(x) = \log(1+x).$$

With $SL(\Sigma_p, \hat{\Sigma}_{\text{Ideal}, m})/SL(\Sigma_p, \hat{\Sigma}_{\text{Ideal}}) \xrightarrow{a.s.} 1$ by lemma S.4,

$$\begin{aligned} EL(\Sigma_p, \hat{\Sigma}_m) &= 1 - \left(\frac{p^{-1} \sum_{i=1}^p h_1(g_i)}{p^{-1} SL(\Sigma_p, \hat{\Sigma}_{\text{Ideal}})} \right. \\ &\quad \left. + \frac{p^{-1} \sum_{i=1}^p h_2(g_i)}{p^{-1} SL(\Sigma_p, \hat{\Sigma}_{\text{Ideal}})} + \frac{p^{-1} SL(\Sigma_p, \hat{\Sigma}_{\text{Ideal}, m})}{p^{-1} SL(\Sigma_p, \hat{\Sigma}_{\text{Ideal}})} \right)^{-1} \\ &\xrightarrow{a.s.} 0, \end{aligned}$$

if we can show further that $p^{-1} \sum_{i=1}^p h_j(g_i) \xrightarrow{a.s.} 0$ for $j = 1, 2$. To this end, by Lemma 1, g_i is almost surely 0 for all i . Hence, in particular, for n large

enough, we have $|g_i| \leq B < 1$ almost surely, so that $h_j(g_i)$ is well defined for all i . This implies that for $j = 1, 2$,

$$p^{-1} \sum_{i=1}^p h_j(g_i) \leq \max_{1 \leq i \leq p} h_j(g_i) = \max_{1 \leq i \leq p} |g_i| |h'_j(\eta_i)| \xrightarrow{a.s.} 0,$$

where η_i lies almost surely in $[-B, B]$ for all i for large enough n , so that the $h'_j(\eta_i)$'s are bounded almost surely for all i . This completes the proof of the theorem. $\square$

Proof of Theorem 6. For the Frobenius loss,

$$\begin{aligned} \|\widehat{\Sigma}_{m,M} - \Sigma_p\|_F^2 &= \left\| \frac{1}{M} \sum_{i=1}^M (\widehat{\Sigma}_m^{(i)} - \Sigma_p) \right\|_F^2 \leq \left(\frac{1}{M} \sum_{i=1}^M \|\widehat{\Sigma}_m^{(i)} - \Sigma_p\|_F \right)^2 \\ &\leq \frac{1}{M} \sum_{i=1}^M \|\widehat{\Sigma}_m^{(i)} - \Sigma_p\|_F^2, \end{aligned}$$

so that

$$\begin{aligned} EL(\Sigma_p, \widehat{\Sigma}_{m,M}) &\leq 1 - \frac{\|\widehat{\Sigma}_{\text{Ideal}} - \Sigma_p\|_F^2}{\frac{1}{M} \sum_{i=1}^M \|\widehat{\Sigma}_m^{(i)} - \Sigma_p\|_F^2} = 1 - \frac{1}{\frac{1}{M} \sum_{i=1}^M \frac{1}{1 - EL(\Sigma_p, \widehat{\Sigma}_m^{(i)})}} \\ &\xrightarrow{a.s.} 0, \end{aligned}$$

since $EL(\Sigma_p, \widehat{\Sigma}_m^{(i)}) \xrightarrow{a.s.} 0$ by Theorem 5. This completes the proof for the Frobenius loss.

For the inverse Stein's loss, note that for any estimator $\widehat{\Sigma}$,

$$SL(\Sigma_p, \widehat{\Sigma}) = \text{tr}\left((\Sigma_p - \widehat{\Sigma})\widehat{\Sigma}^{-1}\right) + \log \det(\widehat{\Sigma}\Sigma_p^{-1}).$$

Hence

$$\begin{aligned} SL(\Sigma_p, \widehat{\Sigma}_{m,M}) &= \frac{1}{M} \sum_{i=1}^M \text{tr}\left((\Sigma_p - \widehat{\Sigma}_m^{(i)})\widehat{\Sigma}_{m,M}^{-1}\right) + \log \det\left(\frac{1}{M} \sum_{i=1}^M \widehat{\Sigma}_m^{(i)} \Sigma_p^{-1}\right) \\ &\leq \frac{1}{M} \sum_{i=1}^M \text{tr}\left((\Sigma_p - \widehat{\Sigma}_m^{(i)})\widehat{\Sigma}_{m,M}^{-1}\right) + \frac{1}{M} \sum_{i=1}^M \log \det(\widehat{\Sigma}_m^{(i)} \Sigma_p^{-1}) \\ &= \frac{1}{M} \sum_{i=1}^M \text{tr}\left((\Sigma_p - \widehat{\Sigma}_m^{(i)})(\widehat{\Sigma}_{m,M}^{-1} - (\widehat{\Sigma}_m^{(i)})^{-1})\right) + \frac{1}{M} \sum_{i=1}^M SL(\Sigma_p, \widehat{\Sigma}_m^{(i)}), \end{aligned}$$

where the second line comes from the convexity of $\log \det(\mathbf{X})$ where $\mathbf{X}$ is positive semi-definite. Then

$$\begin{aligned} EL(\Sigma_p, \hat{\Sigma}_{m,M}) &\leq 1 - \left(\frac{\frac{1}{pM} \sum_{i=1}^M \text{tr} \left((\Sigma_p - \hat{\Sigma}_m^{(i)}) (\hat{\Sigma}_{m,M}^{-1} - (\hat{\Sigma}_m^{(i)})^{-1}) \right)}{p^{-1} SL(\Sigma_p, \hat{\Sigma}_{\text{Ideal}})} \right. \\ &\quad \left. + \frac{1}{M} \sum_{i=1}^M \frac{1}{1 - EL(\Sigma_p, \hat{\Sigma}_m^{(i)})} \right)^{-1} \\ &\xrightarrow{a.s.} 0, \end{aligned}$$

if we can show further that $\frac{1}{pM} \sum_{i=1}^M \text{tr} \left((\Sigma_p - \hat{\Sigma}_m^{(i)}) (\hat{\Sigma}_{m,M}^{-1} - (\hat{\Sigma}_m^{(i)})^{-1}) \right) \xrightarrow{a.s.} 0$, since by Theorem 5 we have $EL(\Sigma_p, \hat{\Sigma}_m^{(i)}) \xrightarrow{a.s.} 0$ for each i , and that in the proof of Lemma S.4 it is proved that $p^{-1} SL(\Sigma_p, \hat{\Sigma}_{\text{Ideal}})$ converges almost surely to a nonzero value when $\Sigma_p \neq \sigma^2 \mathbf{I}$. For this, write the term as $D_1 + D_2$, where

$$D_1 = \frac{1}{pM} \sum_{i=1}^M \text{tr}((\hat{\Sigma}_m^{(i)} - \Sigma_p)(\hat{\Sigma}_m^{(i)})^{-1}), \quad D_2 = \frac{1}{pM} \sum_{i=1}^M \text{tr}((\Sigma_p - \hat{\Sigma}_m^{(i)})\hat{\Sigma}_{m,M}^{-1}).$$

For D_1 , writing $\mathbf{P}_{1i} = (\mathbf{p}_{1i,1}, \dots, \mathbf{p}_{1i,p})$, we can use Lemma 1 to deduce that

$$\begin{aligned} D_1 &= \frac{1}{pM} \sum_{i=1}^M \text{tr}((\text{diag}(\mathbf{P}_{1i}^T \tilde{\Sigma}_2^{(i)} \mathbf{P}_{1i}) - \mathbf{P}_{1i}^T \Sigma_p \mathbf{P}_{1i}) \text{diag}^{-1}(\mathbf{P}_{1i}^T \tilde{\Sigma}_2^{(i)} \mathbf{P}_{1i})) \\ &= \frac{1}{pM} \sum_{i=1}^M \sum_{j=1}^p \frac{\mathbf{p}_{1i,j}^T \tilde{\Sigma}_2^{(i)} \mathbf{p}_{1i,j} - \mathbf{p}_{1i,j}^T \Sigma_p \mathbf{p}_{1i,j}}{\mathbf{p}_{1i,j}^T \tilde{\Sigma}_2^{(i)} \mathbf{p}_{1i,j}} \\ &\leq \frac{1}{M} \sum_{i=1}^M \max_{1 \leq j \leq p} \left| \frac{\mathbf{p}_{1i,j}^T \tilde{\Sigma}_2^{(i)} \mathbf{p}_{1i,j} - \mathbf{p}_{1i,j}^T \Sigma_p \mathbf{p}_{1i,j}}{\mathbf{p}_{1i,j}^T \tilde{\Sigma}_2^{(i)} \mathbf{p}_{1i,j}} \right| \\ &\quad \cdot \left(1 - \max_{1 \leq j \leq p} \left| \frac{\mathbf{p}_{1i,j}^T \tilde{\Sigma}_2^{(i)} \mathbf{p}_{1i,j} - \mathbf{p}_{1i,j}^T \Sigma_p \mathbf{p}_{1i,j}}{\mathbf{p}_{1i,j}^T \tilde{\Sigma}_2^{(i)} \mathbf{p}_{1i,j}} \right| \right)^{-1} \xrightarrow{a.s.} 0. \end{aligned}$$

For D_2 , define $\hat{\Sigma}_{\text{CRC}}^{(i)} = \mathbf{P}_{1i} \left(\frac{1}{M} \sum_{j=1}^M \text{diag}(\mathbf{P}_{1j}^T \tilde{\Sigma}_2^{(j)} \mathbf{P}_{1j}) \right) \mathbf{P}_{1i}$, and define also $\mathbf{B}_i = (\hat{\Sigma}_{\text{CRC}}^{(i)} - \hat{\Sigma}_{m,M})(\hat{\Sigma}_{\text{CRC}}^{(i)})^{-1}$. Then

$$(S.4) \quad \hat{\Sigma}_{m,M}^{-1} = (\hat{\Sigma}_{\text{CRC}}^{(i)})^{-1} (\mathbf{I} - \mathbf{B}_i)^{-1} = (\hat{\Sigma}_{\text{CRC}}^{(i)})^{-1} + (\hat{\Sigma}_{\text{CRC}}^{(i)})^{-1} \sum_{k \geq 1} \mathbf{B}_i^k.$$

Suppose $\max_{1 \leq i, j \leq M} \|\mathbf{P}_{1i} - \mathbf{P}_{1j}\| \xrightarrow{a.s.} 0$, to be proved at the end of this proof. Then the above Neumann's series expansion is valid, since then for large enough n, p with $p/n \rightarrow c > 0$,

$$\begin{aligned}
\|\mathbf{B}_i\| &\leq \frac{1}{M} \left(2 \left\| \sum_{j=1}^M (\mathbf{P}_{1i} - \mathbf{P}_{1j}) \text{diag}(\mathbf{P}_{1j}^T \tilde{\Sigma}_2^{(j)} \mathbf{P}_{1j}) \mathbf{P}_{1i}^T \right\| \right. \\
&\quad \left. + \left\| \sum_{j=1}^M (\mathbf{P}_{1i} - \mathbf{P}_{1j}) \text{diag}(\mathbf{P}_{1j}^T \tilde{\Sigma}_2^{(j)} \mathbf{P}_{1j}) (\mathbf{P}_{1i} - \mathbf{P}_{1j})^T \right\| \right) \\
&\quad \cdot \left(\min_{1 \leq j \leq M} \min_{1 \leq k \leq p} \mathbf{p}_{1j,k}^T \tilde{\Sigma}_2^{(j)} \mathbf{p}_{1j,k} \right)^{-1} \\
&\leq \max_{1 \leq j \leq M} \|\mathbf{P}_{1i} - \mathbf{P}_{1j}\| \max_{1 \leq k \leq p} \mathbf{p}_{1j,k}^T \Sigma_p \mathbf{p}_{1j,k} \left(2 + \|\mathbf{P}_{1i} - \mathbf{P}_{1j}\| \right) \\
&\quad \cdot \left(\min_{1 \leq j \leq M} \min_{1 \leq k \leq p} \mathbf{p}_{1j,k}^T \Sigma_p \mathbf{p}_{1j,k} \right)^{-1} \\
\text{(S.5)} \quad &\leq \max_{1 \leq i, j \leq M} \|\mathbf{P}_{1i} - \mathbf{P}_{1j}\| \|\Sigma_p\| (2 + \|\mathbf{P}_{1i} - \mathbf{P}_{1j}\|) / \lambda_{\min}(\Sigma_p) \xrightarrow{a.s.} 0,
\end{aligned}$$

where the second last line follows from Lemma 1 when n, p is large. With (S.4), $D_2 = D_3 + D_4$, where

$$\begin{aligned}
D_3 &= \frac{1}{pM} \sum_{i=1}^M \text{tr} \left((\Sigma_p - \hat{\Sigma}_m^{(i)}) (\hat{\Sigma}_{\text{CRC}}^{(i)})^{-1} \right), \\
D_4 &= \frac{1}{pM} \sum_{i=1}^M \text{tr} \left((\Sigma_p - \hat{\Sigma}_m^{(i)}) (\hat{\Sigma}_{\text{CRC}}^{(i)})^{-1} \sum_{k \geq 1} \mathbf{B}_i^k \right).
\end{aligned}$$

Similar to the analysis of D_1 , using Lemma 1, we have

$$\begin{aligned}
D_3 &= \frac{1}{pM} \sum_{i=1}^M \sum_{k=1}^p \frac{\mathbf{p}_{1i,k}^T \Sigma_p \mathbf{p}_{1i,k} - \mathbf{p}_{1i,k}^T \tilde{\Sigma}_2^{(i)} \mathbf{p}_{1i,k}}{\frac{1}{M} \sum_{j=1}^M \mathbf{p}_{1j,k}^T \tilde{\Sigma}_2^{(j)} \mathbf{p}_{1j,k}} \\
&\leq \max_{1 \leq i \leq M} \max_{1 \leq k \leq p} \left| \frac{\mathbf{p}_{1i,k}^T \Sigma_p \mathbf{p}_{1i,k} - \mathbf{p}_{1i,k}^T \tilde{\Sigma}_2^{(i)} \mathbf{p}_{1i,k}}{\mathbf{p}_{1i,k}^T \Sigma_p \mathbf{p}_{1i,k}} \right| \cdot \mathbf{p}_{1i,k}^T \Sigma_p \mathbf{p}_{1i,k} \\
&\quad \cdot \left(\min_{1 \leq j \leq M} \min_{1 \leq k \leq p} \left(\frac{\mathbf{p}_{1j,k}^T \tilde{\Sigma}_2^{(j)} \mathbf{p}_{1j,k} - \mathbf{p}_{1j,k}^T \Sigma_p \mathbf{p}_{1j,k}}{\mathbf{p}_{1j,k}^T \Sigma_p \mathbf{p}_{1j,k}} \right) \mathbf{p}_{1j,k}^T \Sigma_p \mathbf{p}_{1j,k} \right. \\
&\quad \left. + \mathbf{p}_{1j,k}^T \Sigma_p \mathbf{p}_{1j,k} \right)^{-1} \xrightarrow{a.s.} 0.
\end{aligned}$$

For D_4 , with $\|\mathbf{B}_i\| \xrightarrow{a.s.} 0$ proved in (S.5) and using Lemma 1, we have

$$\begin{aligned} D_4 &\leq \max_{1 \leq i \leq M} \|\widehat{\Sigma}_m^{(i)} - \Sigma_p\| \cdot \lambda_{\min}^{-1}(\widehat{\Sigma}_{\text{CRC}}^{(i)}) \cdot \sum_{k \geq 1} \|\mathbf{B}_i\|^k \\ &\leq \frac{(\|\Sigma_p\| + \max_{1 \leq i \leq M} \max_{1 \leq k \leq p} \mathbf{p}_{1i,k}^T \widetilde{\Sigma}_2^{(i)} \mathbf{p}_{1i,k}) \|\mathbf{B}_i\|}{\min_{1 \leq i \leq M} \min_{1 \leq k \leq p} \mathbf{p}_{1i,k}^T \widetilde{\Sigma}_2^{(i)} \mathbf{p}_{1i,k} (1 - \|\mathbf{B}_i\|)} \xrightarrow{a.s.} 0. \end{aligned}$$

Hence the proof of the Theorem is complete if we can show further that $\max_{1 \leq i, j \leq M} \|\mathbf{P}_{1i} - \mathbf{P}_{1j}\| \xrightarrow{a.s.} 0$. To this end, using (S.3), we have for any $1 \leq i \leq M$, with $\lambda_{1i,k}$ denoting the k th largest eigenvalue of $\widetilde{\Sigma}_1^{(i)}$,

$$p^{-1} \sum_{k=1}^p g(\mathbf{p}_{1i,k}^T \Sigma_p \mathbf{p}_{1i,k}) \mathbf{1}_{\{\lambda_{1i,k} \leq x\}} \xrightarrow{a.s.} \int_{-\infty}^x g(\delta(\lambda)) dF(\lambda),$$

since $m/n \rightarrow 1$ implies that $p/n, p/m \rightarrow c > 0$. Setting $g \equiv 1$, it is easy to see that $\lambda_{1i,k}$ is almost surely the same as $\lambda_{1j,k}$ for $1 \leq i, j \leq M$ and $1 \leq k \leq p$, as $n, p \rightarrow \infty$. These imply that for $1 \leq i \leq M$, $\mathbf{p}_{1i,k}^T \Sigma_p \mathbf{p}_{1i,k}$ is almost surely the same as $\delta(\lambda_{11,k})$, and hence for $1 \leq i, j \leq M$ and $1 \leq k \leq p$,

$$\mathbf{p}_{1i,k}^T \Sigma_p \mathbf{p}_{1i,k} - \mathbf{p}_{1j,k}^T \Sigma_p \mathbf{p}_{1j,k} \xrightarrow{a.s.} 0.$$

But $\mathbf{p}_{1i,k}^T \Sigma_p \mathbf{p}_{1i,k} - \mathbf{p}_{1j,k}^T \Sigma_p \mathbf{p}_{1j,k} = (\mathbf{p}_{1i,k} - \mathbf{p}_{1j,k})^T \Sigma_p (\mathbf{p}_{1i,k} + \mathbf{p}_{1j,k})$, hence if $\mathbf{p}_{1i,k}$ or $\mathbf{p}_{1j,k}$ are not eigenvectors of Σ_p asymptotically (this excludes the case $\Sigma_p = \sigma^2 \mathbf{I}$, which by assumption is not the case), then there exists a constant $a > 0$ such that

$$(\mathbf{p}_{1i,k} - \mathbf{p}_{1j,k})^T \Sigma_p (\mathbf{p}_{1i,k} + \mathbf{p}_{1j,k}) = \|\mathbf{p}_{1i,k} - \mathbf{p}_{1j,k}\| \cdot a \|\mathbf{p}_{1i,k} + \mathbf{p}_{1j,k}\| \xrightarrow{a.s.} 0,$$

implying that for all $1 \leq i, j \leq M$, $1 \leq k \leq p$, $\|\mathbf{p}_{1i,k} \pm \mathbf{p}_{1j,k}\| \xrightarrow{a.s.} 0$. With the correct orientation chosen, we must then have $\max_{1 \leq i, j \leq M} \|\mathbf{P}_i - \mathbf{P}_j\| \xrightarrow{a.s.} 0$.

Finally, if both $\mathbf{p}_{1i,k}$ and $\mathbf{p}_{1j,k}$ are asymptotic eigenvectors of Σ_p for some $1 \leq k \leq p$, then assuming $\|\Sigma_p \mathbf{p}_{1i,k} - \lambda \mathbf{p}_{1i,k}\| \xrightarrow{a.s.} 0$ and $\|\Sigma_p \mathbf{p}_{1j,k} - \gamma \mathbf{p}_{1j,k}\| \xrightarrow{a.s.} 0$,

$$\begin{aligned} (\mathbf{p}_{1i,k} - \mathbf{p}_{1j,k})^T \Sigma_p (\mathbf{p}_{1i,k} + \mathbf{p}_{1j,k}) &= (\mathbf{p}_{1i,k} - \mathbf{p}_{1j,k})^T (\lambda \mathbf{p}_{1i,k} + \gamma \mathbf{p}_{1j,k}) \\ &\quad + (\mathbf{p}_{1i,k} - \mathbf{p}_{1j,k})^T (\Sigma_p \mathbf{p}_{1i,k} - \lambda \mathbf{p}_{1i,k} \\ &\quad + \Sigma_p \mathbf{p}_{1j,k} - \gamma \mathbf{p}_{1j,k}) \\ &= (\lambda - \gamma)(1 - \mathbf{p}_{1i,k}^T \mathbf{p}_{1j,k}) + o(1), \end{aligned}$$

so that if $\lambda \neq \gamma$, we must have $\|\mathbf{p}_{1i,k} - \mathbf{p}_{1j,k}\| \xrightarrow{a.s.} 0$, which is what we want. If $\lambda = \gamma$, then $\mathbf{p}_{1i,k}$ and $\mathbf{p}_{1j,k}$ are asymptotically spanning the same eigenspace, hence $\max_{1 \leq i, j \leq M} \|\mathbf{P}_i - \mathbf{P}_j\| \xrightarrow{a.s.} 0$ still holds. This completes the proof of the theorem. $\square$

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